

ASAD HUSSAIN

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PROFESSIONAL SUMMARY

Final-year Electrical Engineering student at NUST working across applied machine learning and embedded/RF systems engineering backend/AI work at FlyRank AI, ML projects at CodeAlpha, and deep learning/LLM/RAG training via AtomCamp on one side; DroneGuard, a two-board RF detection and jamming system, plus embedded firmware and IoT telemetry from prior internships on the other. Comfortable moving between training models and building the circuits and firmware that feed them real-world data.

TECHNICAL SKILLS

Programming: Python · C++/Embedded C · JavaScript · SQL · MATLAB/Simulink · Flask

ML / AI: Scikit-learn · TensorFlow · PyTorch · LangChain · RAG · LLM Fine-tuning · FastAPI · Docker

Hardware & RF: SDR · STM32 · Arduino · Raspberry Pi · PCB Design (Proteus) · Antenna Design

Tools: Git · REST APIs · SQL Server · Google Cloud · Weights & Biases · AutoCAD, Altium Designer

AI/ML EXPERIENCE & TRAINING

FlyRank AI | *Backend AI Engineering Intern (Remote) Chicago, USA*

Jun – Aug 2026

- Completed a backend/AI engineering practicum: built an authentication-protected CRUD API with a connected database, containerized the stack with Docker, and automated a data-scraping/workflow pipeline.
- Applied prompt-engineering and LLM-tooling fundamentals to search-intelligence tasks (baseline scoring and top-10 review pipeline) across 21 practical assignments and 19 workshops on AI tooling and backend development.

CodeAlpha | *Machine Learning Intern (Remote)*

May – Jun 2026

- Built a credit-scoring classification model (Logistic Regression, Decision Trees, Random Forest) with feature engineering from financial history data, evaluated using Precision, Recall, F1-score, and ROC-AUC.
- Built a CNN-based handwritten character recognition model (MNIST/EMNIST) and a disease-prediction classifier on structured medical data (symptoms, age, lab results) using SVM, Logistic Regression, and Random Forest.

AtomCamp — AI Bootcamp

- Completed training in deep learning, NLP/LLMs, n8n, generative AI & RAG (LangChain agent development), and MLOps/deployment (Docker, FastAPI, CI/CD, Google Cloud).

WORK EXPERIENCE

Technical Intern | *National Electric Power Regulatory Authority (NEPRA), Islamabad*

Jul – Aug 2026

- Studied NEPRA's end-to-end regulatory flow (license → tariff → sale) across Generation, Transmission, Distribution, and Supply licenses, including the three generation tariff mechanisms.
- Analyzed NGC's transmission network data, 233 grid-station transformers (116 above 80% rated capacity) and NEPRA's State of the Industry Report, covering 111,466 GWh of national consumption.

Engineering Intern, Full-Stack Systems | *Research & Indigenous Development Centre (RDC), HIT*

Jun – Aug 2025

- Integrated core hardware (LilyGO T-SIM7670E, T3-S3 GPS, RDM6300 RFID) with C/C++ state machines for GPS capture, RFID authentication, and MQTT telemetry over cellular GPRS.
- Built an asynchronous Python Flask backend with multithreading to process analytics workloads without blocking incoming MQTT streams.

FINAL YEAR PROJECT

DroneGuard — Anti-Drone Surveillance System

2025 – 2026

SDR · GNU Radio · Raspberry Pi 4 · STM32H743 · MATLAB · Python

- Designed a two-board anti-drone system: a four-band RF detection front-end (433 MHz–2.4 GHz) and a jamming board built around an ADF4351 PLL synthesiser.
- Built a real-time signal-classification pipeline (RTL-SDR + GNU Radio) identifying FHSS, OFDM, and FSK drone protocols from live IQ samples, plus a MATLAB front-end for live spectrum and telemetry.

EDUCATION

National University of Sciences and Technology (NUST)

2023 – 2027

B.E. Electrical Engineering — Coursework: Analog Electronics, DSP, Control Systems, Communication Systems, Digital Logic Design, Embedded Systems, Signals & Systems, Instrumentation, OOP, Microprocessor Systems

ACADEMIC PROJECTS

Hands-on projects across core EE disciplines: Butterworth Active LPF (op-amp) · Multi-Stage Audio Amplifier · EEG Cleaning MATLAB GUI · PID DC Motor Controller (Arduino) · 5-Band Graphic Equaliser (Simulink) · 4-Bit ALU · FM Receiver (TDA7000) · Multi-Range Voltmeter · OOP Shopping Cart (C++) · Water Level Indicator (Proteus) · Current Transformer

CERTIFICATIONS, ACTIVITIES & ACHIEVEMENTS

MATLAB Fundamentals (MathWorks) · PCB Design (Proteus & SPICE) · Technical Lead, NUST Robotics Society · Highest Achiever, Matriculation (SSC) · 1st Place Robotics, Voltfest · NMUN 2024 Delegate

PORTFOLIO

[Asad Hussain | Electrical Embedded Engineer Portfolio](#) — full-stack portfolio showcasing embedded, RF/SDR, and ML/AI projects.